

A Research Paper: Multi-Dimensional Assessment Frameworks for Mathematical Proficiency

Prepared for the MathPivot Development Team

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Executive Summary

This paper introduces three research-backed assessment frameworks—**Skill Progression**, **Cognitive Depth**, and **Real-World Relevance**—designed to replace traditional “easy/medium/hard” classifications in math education. These frameworks enable granular, standards-aligned assessment design by categorizing questions along three orthogonal axes:

- Developmental mastery (**Skill Progression**),
- Intellectual demand (**Cognitive Depth**),
- Contextual authenticity (**Real-World Relevance**).

We demonstrate their application using two atomic concepts:

- **Systems of Linear Equations (A-REI.5)**
- **Slope as Rate of Change (NC.M1.F-IF.6)**

Finally, we detail their centrality to MathPivot’s adaptive assessment engine.

Framework 1: Skill Progression

Definition: Tracks a student’s journey from basic skill acquisition to strategic application.

Level	Atomic Concept 1: Systems of Linear Equations (A-REI.5)	Atomic Concept 2: Slope as Rate of Change (NC.M1.F-IF.6)
Foundational (Define/Verify)	Task: "Define a system of linear equations. Verify if (1, 4) solves: $y = 3x + 1$ and $y = -x + 5$." Expected Answer: "System: Two or more linear equations with shared variables. Verification: $4 = 3(1) + 1 \rightarrow \text{True}$; $4 = -1 + 5 \rightarrow \text{True}$. Solution valid."	Task: "Define slope. Calculate slope between (2, 8) and (5, 17)." Expected Answer: "Slope = vertical change / horizontal change. $m = \frac{17-8}{5-2} = \frac{9}{3} = 3$."
Applied (Solve Standard Problems)	Task: "Solve algebraically: $2x + y = 9$ and $3x - y = 1$." Expected Answer: "Add equations: $5x = 10 \rightarrow x = 2$. Substitute: $2(2) + y = 9 \rightarrow y = 5$. Solution: (2, 5)." 	Task: "A car travels 180 miles in 3 hours. Calculate speed (slope) and interpret it." Expected Answer: " $m = \frac{180}{3} = 60$ mph. Interpretation: Speed is 60 miles per hour."
Strategic (Design/Justify)	Task: "Create a system with no solution. Explain why." Expected Answer: "System: $y = 4x + 2$ and $y = 4x - 3$. Explanation: Parallel lines (same slope, different intercepts) never intersect."	Task: "A water tank drains 30 gallons in 10 minutes. Express slope as a unit rate. Explain how slope changes if time is in seconds." Expected Answer: "Slope = -3 gal/min (draining). If time in seconds: $m = \frac{-30}{600} = -0.05$ gal/sec. Units alter the numeric value."

Framework 2: Cognitive Depth

Definition: Measures complexity of thinking required.

Level	Atomic Concept 1: Systems of Linear Equations (A-REI.5)	Atomic Concept 2: Slope as Rate of Change (NC.M1.F-IF.6)
Recall/Compute (Basic Facts)	Task: "Graph shows lines intersecting at (3, -1). Identify the solution." Expected Answer: "(3, -1)." 	Task: "Find slope from graph: Line passes through (0, 0) and (4, 12)." Expected Answer: $m = \frac{12-0}{4-0} = 3.$
Analyze/Interpret (Relational Reasoning)	Task: "Without solving, explain why $y = 2x + 3$ and $4x - 2y = -6$ have infinite solutions." Expected Answer: "Second equation: $4x - 2y = -6 \rightarrow -2y = -4x - 6 \rightarrow y = 2x + 3$. Identical lines $\rightarrow$ infinite solutions." 	Task: "Distance-time graph for two bikes: Bike A: (0,0) to (2,10); Bike B: (0,0) to (4,12). Which is faster? Use slope." Expected Answer: "Bike A slope: $\frac{10}{2} = 5$ mph. Bike B slope: $\frac{12}{4} = 3$ mph. Bike A is faster."
Evaluate/Create (Critical Judgment)	Task: "Critique: 'If slopes differ, a system has one solution.' Provide counterexamples." Expected Answer: "False. Counterexample 1 (no solution): $y = 3x$ and $y = 3x + 4$ (parallel). Counterexample 2 (infinite solutions): $2x + y = 4$ and $4x + 2y = 8$ (same line)." 	Task: "Design a scenario where slope = -2.5 has meaning. Explain why the sign matters." Expected Answer: "Scenario: A submarine descends 250 feet in 100 seconds. Slope = -2.5 ft/sec. Negative sign indicates descent."

Framework 3: Real-World Relevance

Definition: Evaluates contextualization from abstract to authentic modeling.

Level	Atomic Concept 1: Systems of Linear Equations (A-REI.5)	Atomic Concept 2: Slope as Rate of Change (NC.M1.F-IF.6)
Procedural (No Context)	Task: "Solve: $5x - y = 11$ and $x + y = 1$." Expected Answer: "Add equations: $6x = 12 \rightarrow x = 2$. Substitute: $2 + y = 1 \rightarrow y = -1$."	Task: "Calculate slope between $(-1, 4)$ and $(3, -8)$." Expected Answer: " $m = \frac{-8-4}{3-(-1)} = \frac{-12}{4} = -3$."
Contextual (Simple Setup)	Task: "Apples cost \$2, oranges cost \$3. Bought 10 fruits for \$24. Write/solve a system." Expected Answer: "System: $a + o = 10$ and $2a + 3o = 24$. Solution: $a = 6, o = 4$."	Task: "A tree grew from 6 ft to 9 ft in 18 months. Find growth rate (slope)." Expected Answer: " $m = \frac{9-6}{18} = \frac{3}{18} = 0.167$ ft/month."
Scenario-Based (Complex Modeling)	Task: "A café sells coffee (\$3) and tea (\$2). Day 1: 40 drinks, \$104 revenue. Day 2: 10% discount on coffee, new revenue \$98 for 40 drinks. How many coffees sold at discount?" Expected Answer: "Day 1: $c + t = 40$; $3c + 2t = 104$. Day 2: $c + t = 40$; $2.7c + 2t = 98$ (10% off coffee $\rightarrow$ \$2.70). Solve: $c = 20, t = 20$. 20 coffees sold at discount."	Task: "Electricity costs: \$0.15/kWh base rate + \$10 service fee. Graph cost vs. kWh (0-200 kWh). Calculate slope. How does slope change if fee increases to \$15?" Expected Answer: "Equation: Cost = $0.15k + 10$. Slope = 0.15 (\$/kWh). If fee = \$15, slope remains 0.15 (slope is rate, fee is y-intercept)."

Conclusion for MathPivot

These three orthogonal frameworks—Skill Progression, Cognitive Depth, and Real-World Relevance—provide a robust, multi-dimensional model for assessing mathematical proficiency. By moving beyond one-dimensional difficulty labels, MathPivot’s adaptive engine can precisely diagnose student understanding, target specific developmental zones, and generate personalized learning pathways that build both procedural fluency and conceptual mastery. This granular approach ensures assessments are aligned with educational standards while being responsive to individual learner needs.